

Cheat Sheet

$$\cos(at) \cos(bt) = \frac{1}{2}(\cos((a+b)t) + \cos((a-b)t))$$

Fourier Series

Trig

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t + b_n \sin n\omega_0 t$$

Where:

$$T_0 = \frac{2\pi}{\omega_0}$$

a_0 = the average

$$a_n = \frac{2}{T_0} \int_{t_1}^{t_1+T_0} x(t) \cos n\omega_0 t \, dt$$

$$b_n = \frac{2}{T_0} \int_{t_1}^{t_1+T_0} x(t) \sin n\omega_0 t \, dt$$

Compact

$$x(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n)$$

Where:

$$C_0 = a_0$$

$$C_n = \sqrt{a_n^2 + b_n^2}$$

$$\theta_n = \arctan(-b_n, a_n)$$

Exponential

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}$$

Where:

$$D_n = \frac{1}{T_0} \int_{T_0} x(t) e^{-jn\omega_0 t} \, dt$$

Power

$$P_x = C_0 + \frac{1}{2} \sum_{n=1}^{\infty} C_n^2 = \sum_{n=-\infty}^{\infty} |D_n|^2$$

Band Width

$$BW = \omega_{max} - \omega_{min}$$

modulation property

Fourier Transform

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} \, dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} \, d\omega$$

- Linear
- $x^*(t) \iff X^*(-\omega)$
- $2\pi x(-\omega) \iff X(t)$
- $x(at) \iff \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$
- $x(t - t_0) \iff X(\omega) e^{-j\omega t_0}$
- $x(t) e^{j\omega_0 t} \iff X(\omega - \omega_0)$
- $x_1(t) * x_2(t) \iff X_1(\omega) X_2(\omega)$
- $x_1(t) x_2(t) \iff \frac{1}{2\pi} X_1(\omega) * X_2(\omega)$
- $\frac{d^n x(t)}{dt^n} \iff (j\omega)^n X(\omega)$
- $\int_{-\infty}^t x(u) \, du \iff \frac{X(\omega)}{j\omega} + \pi X(0) \delta(\omega)$

Common Transforms

$x(t)$	$X(\omega)$
$e^{-at}u(t)$	$\frac{1}{a+j\omega}$
$e^{at}u(-t)$	$\frac{1}{a-j\omega}$
$e^{-a t }$	$\frac{2a}{a^2+\omega^2}$
$te^{-at}u(t)$	$\frac{1}{(a+j\omega)^2}$
$t^n e^{-at}u(t)$	$\frac{n!}{(a+j\omega)^{n+1}}$
$\delta(t)$	1
1	$2\pi\delta(\omega)$
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
$\cos \omega_0 t$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
$\sin \omega_0 t$	$j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$
$u(t)$	$\pi\delta(\omega) + \frac{1}{j\omega}$
$\text{sgn}(t)$	$\frac{2}{j\omega}$
$\cos \omega_0 t u(t)$	$\frac{\pi}{2}[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] + \frac{j\omega}{\omega_0^2 - \omega^2}$
$\sin \omega_0 t u(t)$	$\frac{\pi}{2j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)] + \frac{\omega_0}{\omega_0^2 - \omega^2}$
$e^{-at} \sin \omega_0 t u(t)$	$\frac{\omega_0}{(a+j\omega)^2 + \omega_0^2}$
$e^{-at} \cos \omega_0 t u(t)$	$\frac{a+j\omega}{(a+j\omega)^2 + \omega_0^2}$
$\text{rect}\left(\frac{t}{\tau}\right)$	$\tau \text{sinc}\left(\frac{\omega\tau}{2}\right)$
$\frac{W}{\pi} \text{sinc}(Wt)$	$\text{rect}\left(\frac{\omega}{2W}\right)$
$\Delta\left(\frac{t}{\tau}\right)$	$\frac{\tau}{2} \text{sinc}^2\left(\frac{\omega\tau}{4}\right)$
$\frac{W}{2\pi} \text{sinc}^2\left(\frac{Wt}{2}\right)$	$\Delta\left(\frac{\omega}{2W}\right)$
$\sum_{n=-\infty}^{\infty} \delta(t - nT)$	$\omega_0 \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0)$
$e^{-t^2/2\sigma^2}$	$\sigma\sqrt{2\pi}e^{-\sigma^2\omega^2/2}$